

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.Santhosh kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Santhosh_____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.
Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment.
Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments.
Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Seasonal sales traffic spike – Auto-scaling and Load Balancing

The number of application servers when traffic rises and reduces them when traffic falls. Load balancing distributes customer requests among all available servers so that no single server becomes overloaded.

For example, during a festival sale on an e-commerce website such as Flipkart, millions of users may browse products and make payments at the same time. Auto-scaling launches additional

servers within minutes, while the load balancer sends each user request to the least busy server. If one server fails, the load balancer redirects traffic to healthy servers, reducing downtime. This improves user experience by providing faster page loading. A company faces unpredictable traffic during seasonal sales. In this situation, cloud auto-scaling automatically increases, smooth checkout, and fewer payment failures.

Monitoring tools such as AWS CloudWatch or Azure Monitor continuously track CPU usage, memory, request count, and response time. When thresholds are exceeded, they trigger scaling actions and alert administrators.

Result: The company handles peak traffic efficiently, avoids crashes, and maintains customer satisfaction during high-demand sales events.

2. Secure cloud backup storage – Durability and Cost

The organization wants to store a large amount of backup data securely. Cloud storage services such as Amazon S3, Azure Blob Storage, or Google Cloud Storage keep multiple copies of data across different disks and data centers. This provides high durability and redundancy even if hardware fails.

Sensitive backup data is protected using encryption. Data is encrypted while being transferred and while stored in the cloud. Access control ensures that only authorized administrators or backup systems can read or restore the data.

Compared with traditional storage, the company does not need to purchase storage hardware, maintain backup servers, or pay for power and cooling. It pays only for the storage it actually uses, making cloud backup more economical.

A disaster recovery example is a company whose main data center is damaged by fire or ransomware. The latest cloud backups are restored to a new environment, allowing business operations to continue quickly.

Result: Cloud backup provides secure, durable, scalable, and cost-effective protection for critical business data.

3. Global mobile app deployment – CDN and Edge Servers

The mobile app company plans to serve users worldwide. A Content Delivery Network (CDN) stores copies of app content such as images, videos, updates, and API responses on servers located in many countries.

When a user in Chennai opens the app, content is delivered from a nearby edge server instead of a distant server in another country. This reduces latency, buffering, and loading time.

Cloud providers support global availability by operating multiple regions and edge locations around the world. Traffic is automatically routed to the nearest available location.

In a real-time streaming application such as YouTube or a gaming application such as PUBG, edge servers deliver video segments or game assets locally. Players experience lower lag, and viewers experience smoother streaming.

Result: CDN and edge servers improve performance, reduce network delay, and provide a consistent user experience across different regions.

4. Financial institution – Compliance and Security

A financial institution must meet strict security and regulatory requirements. Cloud providers support compliance by offering certified infrastructure, audit reports, and security controls that help organizations follow standards such as PCI-DSS, ISO 27001, and local banking regulations.

Logging and monitoring record every login, transaction, and configuration change. Security teams can detect suspicious activities and investigate incidents. Encryption protects customer account details and transaction data both during transmission and while stored.

To satisfy data sovereignty requirements, the bank can choose a cloud region located within its country so that customer data does not leave the required jurisdiction.

For example, during an online money transfer, the transaction is encrypted, access is restricted to authorized banking services, and logs are stored for auditing.

Result: The institution achieves secure cloud usage, regulatory compliance, traceability of transactions, and protection of sensitive financial information.\

5.Containerization and Kubernetes – Modern Application Deployment

The development team wants faster and more consistent deployments. Containers package the application with all required libraries, making them lightweight and quick to start. Unlike virtual machines, containers share the host operating system and use fewer resources.

In a microservices-based application, services such as user authentication, product catalog, shopping cart, and payment run in separate containers.

Kubernetes simplifies deployment by automatically scheduling containers, scaling them during high traffic, restarting failed containers, and performing rolling updates without downtime.

Portability is a major advantage. The same container image can run on AWS, Azure, Google Cloud, or on-premises servers without modification.

For example, an e-commerce company can scale only the payment service during checkout peaks while keeping other services unchanged.

Result: Containerization and Kubernetes provide efficient resource usage, easy scaling, high availability, and portability across different cloud environments.